Concrete Mix Design Optimization Using a Multi-objective Cuckoo Search Algorithm

Sriman Pankaj Boindala and Vasan Arunachalam

Abstract Maximizing Compressive strength and Minimizing the cost has always been the objectives for selection of mix design, but with the increase in recent interests in producing eco-friendly materials, another objective has been added to the above two, making it a complex multi-objective problem. This paper presents a way to solve this multi-objective optimizing problem in a more statistical way, optimizing concrete mix designs to obtain a high-strength, low carbon emission and economical mix design using recently developed nature-inspired algorithm namely Cuckoo Search (CS) algorithm. The algorithm was written in MATLAB and the results obtained consist of mix designs having strengths varying from 20 MPa to 90 MPa and their corresponding carbon emissions from 360 to 500 kg of CO_2 Emissions for 1 m^{-3} of concrete. These results were compared with the results obtained with NSGAII and was concluded that MOCS performed better than NSGAII for this application producing wide spread pareto front.

Keywords Concrete-Mix design · Optimization · Multi-objective cuckoo search · Carbon emissions · Compressive strength

1 Introduction

Concrete is a majorly used material in the construction industry. Finding out best mix proportions to obtain highest compressive strength is a challenging task. With the discovery of other cementitious materials, this task has become much more complicated due to increase in search space (i.e. availability of multiple materials for proportionating). For tackling this problem researchers have focused on developing strength predictive models for various types of concrete [1–7]. With the increase in global awareness towards the need for environmental protection, researchers have started

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focusing on developing eco-friendly concrete mixes having materials with low carbon emissions [8–10]. As a reduction in carbon emissions and increase in compressive strength has been to be slightly contradictory, a need to find out a tool/technique to find the best mixes which can maximize the compressive strength, minimize the eco-index or carbon emissions and minimize the cost per batch is essential. To solve this multi-objective problem, a cuckoo search based multi-objective optimization algorithm [11] has been used in this current study. This kind of multi-objective problems to optimize mix design was done in the past considering fresh-properties of concrete also [12].

2 Optimization Using the Cuckoo Search

Cuckoo Search (CS) is one of the latest meta-heuristic algorithms developed based on the natural behaviour of the cuckoos. It uses a levy-flight based search walk instead of normal random walk [13]. This algorithm was developed by Yang et al. [13], as it is a recent development not much work has been done using this algorithm. Comparative analysis of this algorithm with other meta-heuristic algorithms like PSO, FA, showed that CS algorithm is performing better [14]. After the proposal of this cuckoo search, many modifications have been done to this algorithm and were published. Modified cuckoo search (MCS) [15], Improved cuckoo search [16] are few examples of cuckoo search-based algorithms. Most of the engineering-based problems like spring design and welded beam design are solved by using cuckoo search algorithm [13, 14, 17] are few examples for design application.

2.1 Cuckoo Search Algorithm (Single Objective): [13]

Step 1: Formulate the Objective function $f(v)$, $v = (v_1, v_2, v_3, v_4, \ldots\ldots v_d)$ where f is the function and v is the variable set (nest).
Step 2: Decide upon a number of nests to be considered (in general 10–20 times the number of variables).
Step-3: Generate the values for variables within the bounds for all the nests.
Step-4: Evaluate all the nests and find the best nest.
Step-5: Use this best nest to guide the levy flight step size.
Step-6: Evaluate the new nests produced in step 5
Step-7: Update the best nest
Step-8: replace the nests with a probability P. $P \sim (0–1)$.
Step-9: Evaluate all the nests and update the best nest.
Step-10: Repeat the steps 5, 6, 7, 8, 9 iteratively.
Step-11: After several iterations, the nests converge to a particular solution.

2.2 Cuckoo Search Algorithm (Multi-objective) [11]

Step 1: Formulate the Objective functions $f(v)$, $v = (v_1, v_2, v_3, v_4, \ldots v_d)$ where f is the function and v is the variable set (nest).
Step 2: Decide upon a number of nests N to be considered (in general 10–20 times the number of variables).
Step-3: Generate the values for variables within the bounds for all the nests.
Step-4: Evaluate the nest for all the objective functions
Step-5: Sort and rank the nests using Non-dominated Sorting Algorithm (NDSA) [18]
Step-6: Calculate the crowding number of each nest in the particular Pareto front
Step-7: Leader nest is the nest with a least crowding number in the first Pareto front. [11]
Step-8: Use this leader nest to guide the levy flight step size. and obtain a new set of nests.
Step-9: Combine the old and new nests and again sort them using NDSA and finalize the new nests taking top N nests.
Step-10: Replace the nests with a probability P. $P \sim (0–1)$.
Step-11: Evaluate all the nests and update the leader nest.
Step-12: Repeat steps 4, 5, 6, 7, 8, 9, 10, 11 for a significant number of iterations to obtain the best nests.

3 NSGAII

Nondomination Sorting Genetic Algorithm (NSGA) was developed by Srinivas and Deb, 1995 [19]. This algorithm is only for multi-objective problems and uses Genetic Algorithm (GA) for finding best pareto front. Most of the algorithm mechanism is similar to GA, except the selection procedure. Though this NSGA was revolutionary during that time, but the complexity of the algorithm reduced its popularity. It lacked elitism and the optimal value of sharing parameter σ_{share} was difficult to predict. To overcome these a second version of NSGA namely NSGAII was developed. It mainly focused on developing the selection process. The new NSGAII uses crowding distance to rank the population. This reduced the computational time and was not dependent on the sharing parameter. The pseudo code of NSGA II is shown below

Step-1: Formulate the Objective functions $f_1(p)$ and $f_2(p)$, $p = (p_1, p_2, p_3, p_4, \ldots p_d)$ where f is the function and p is the variable set (population).
Step-2: Decide upon a population size P to be considered (in general 10–20 times the number of variables)
Step-3: Generate the values for variables within the bounds for each population.
Step-4: Evaluate the nest for all the objective functions

Step-5: Sort and rank the nests using Non-dominated Sorting Algorithm (NDSA) [18]
Step-6: Calculate the crowding distance of each nest in the particular Pareto front
Step-7: Generate offspring population:

Binary tournament selection
Crossover and mutation

Step-8: Combine the old and new nests and again sort them using NDSA and finalize the elite population taking top P population.
Step-9: Repeat steps 4, 5, 6, 7, 8 for a significant number of iterations to obtain the most elite population.

4 Mathematical Formulation of the Problem

As mentioned earlier finding out best mix design considering compressive strength, carbon emissions as well as cost is highly essential. To address this the following steps are done:

Step-1: The First task is to obtain the data sets required for establishing a relationship between mix proportions as compressive strength. Various literature of Self-compacting Concrete (SCC) blended with fly ash and GGBS were surveyed and a set of 285 mix proportions of SCC mixes were collected. (for dataset authors can be contacted)

Step-2: For the establishment of relation we considered two types of regression (1) Multiple linear regression Eqs. (1) and (2) Non-linear regression with decimal coefficients Eq. (2).

$$\begin{aligned}\text{Compressive strength} &= a_0 + a_1 * \text{Cement} + a_2 * \text{FlyAsh} \\ &+ a_3 * \text{GGBS} + a_4 * \text{Coarse Aggregate} \\ &+ a_5 * \text{Fine Aggregate} + a_6 * \text{Superplaticiser} \\ &+ a_7 * \text{Water}\end{aligned} \tag{1}$$

$$\begin{aligned}\text{Compressive strength} &= b_0 + b_1 * \text{Cement}^{n_1} + b_2 * \text{FlyAsh}^{n_2} \\ &+ b_3 * \text{GGBS}^{n_3} + b_4 * \text{CoarseAggregate}^{n_4} \\ &+ b_5 * \text{FineAggregate}^{n_5} + b_6 * \text{SuperPlasticizer}^{n_6} \\ &+ b_7 * \text{Water}^{n_7}\end{aligned} \tag{2}$$

The ranges for the coefficients of linear and non-linear regression were assumed to be from -1000 to 1000 and the ranges for exponents of non-linear regression were assumed to be -5 to 5. For finding the coefficients single objective cuckoo search has been used objective being maximizing R^2 between observed versus simulated data

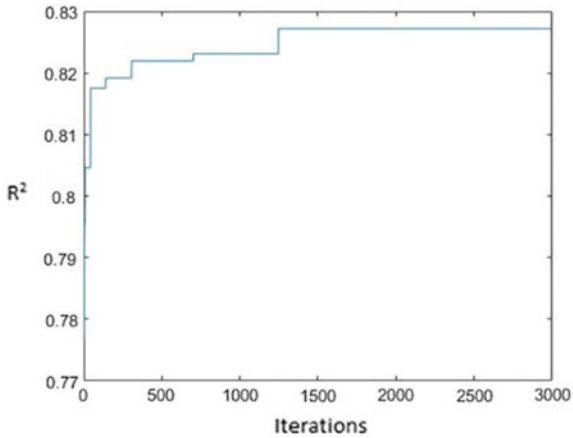

Fig. 1 Graph between R^2 and Iterations obtained after 3000 iterations for establishment of Linear relationship between mix proportions and compressive strength using single objective cuckoo search

and the variables being the coefficients. Among, these two models, MLR showed better results with R^{-2} value to be around 8.26. The increase in R^2 with the number of iterations is shown in Fig. 1. The final relationship obtained is

$$\begin{aligned}\text{Compressive Strength} = &-145.4 + 0.1849 * \text{Cement} + 0.118 * \text{FlyAsh} \\ &+ 0.169 * \text{GGBS} + 0.0623 * \text{Coarse Aggregate} + 0.06972 * \text{Fine Aggregate} \\ &- 0.176 * \text{Superplaticiser} + 0.00281 * \text{Water}\end{aligned} \tag{3}$$

For finding out carbon emissions and cost, a linear relationship is established based to data obtained from fact sheets and other research papers [8, 20–22].

$$\begin{aligned}CO_2 \text{ emmisions} = &\ 0.913 * \text{Cement} + 0.004 * \text{FlyAsh} + 0.067 * \text{GGBS} \\ &+ 0.005 * \text{Coarse Aggregate} + 0.005 * \text{Fine Aggregate} \\ &+ 0.01 * \text{Superplaticiser} + 0.001 * \text{Water}\end{aligned} \tag{4}$$

$$\begin{aligned}\text{Cost} = &\ 8 * \text{Cement} + 0.75 * \text{FlyAsh} + 3.5 * \text{GGBS} \\ &+ 0.85 * \text{Coarse Aggregate} + 3.5 * \text{Fine Aggregate}\end{aligned} \tag{5}$$

After finalizing the objectives, we need to identify our constraints. The constraints are directly based on EFNARC guidelines, since the mixes belong to SCC mixes data set. [23–25]. The constraints are implemented based on exterior penalty function approach. Based on the type of our objective (maximizing for Compressive strength and minimizing cost) we can add or subtract the penalty to alter the objective function value.

Step-3: After obtaining the objective functions, for optimizing these three functions a MATLAB code for Multi-objective Cuckoo Search (MOCS) is been formulated and used. For validating the code written, benchmark problem Viennet Function is used (Fig. 2).

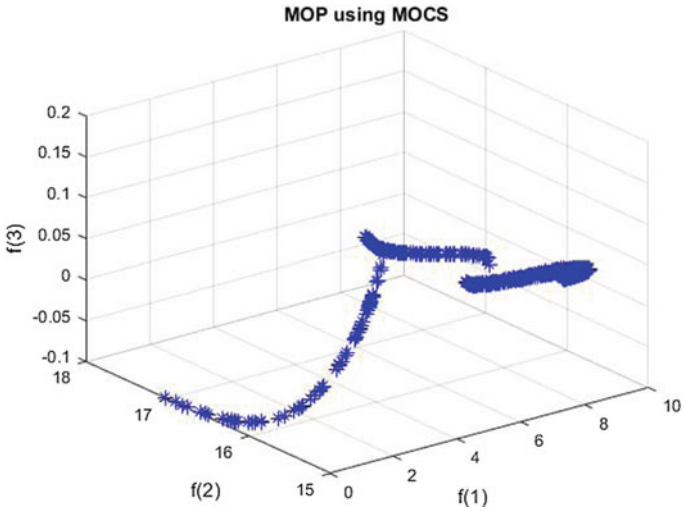

Fig. 2 Graph obtained for the Viennet test function using the developed code

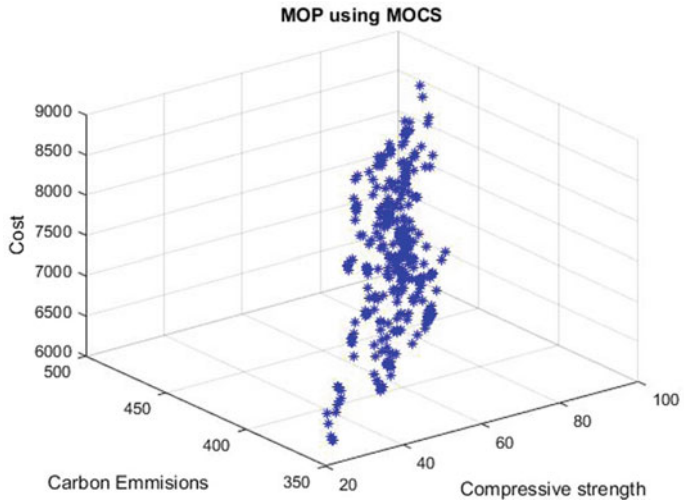

Fig. 3 Result obtained after 1000 iterations using multi-objective cuckoo search algorithm

5 Results

The above formulated mathematical model is solved using MOCS and compared with the results obtained from NSGAII (as shown in Fig. 4). NSGAII is taken for comparison because it is known to be one of the best performing benchmarks for

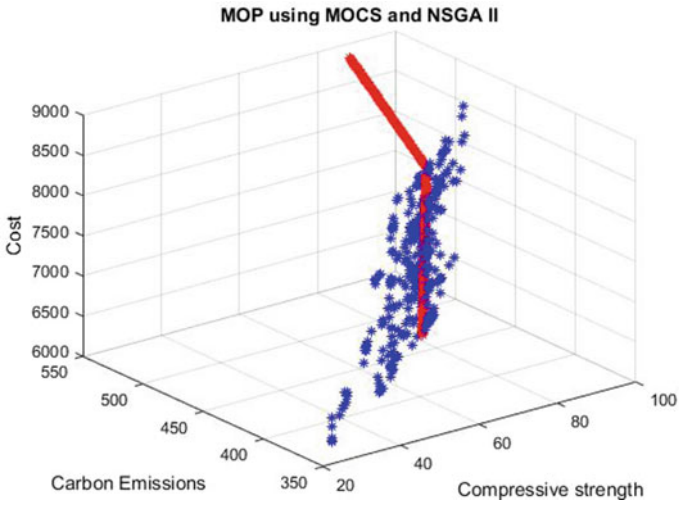

Fig. 4 Comparison of NSGA-II (Red) and multi-objective cuckoo search (Blue)

solving multi objective optimization algorithms. For the establishment of the Multiple linear relationships and Non-linear relationship between mix-proportions and compressive strength, the single objective cuckoo search is used objective being maximizing R^2 value. The final values of R^2 obtained after 3000 iterations are 0.82 and 0.78 respectively. The improvement in R^2 values with increase in the iterations is depicted in Fig. 1. After running the program for 1000 iterations using 350 nests as initial population with beta value 3/2 and alpha levy 0.01 in an I-5 3rd generation 2.5 GHz computer with 8 Gb ram, spread of mix designs (Fig. 3) was obtained with compressive strengths ranging from 20 to 90 MPa and carbon emissions of 360–500 kg of CO_2/m^3 of concrete.

6 Conclusions

- For the establishment of non-linear relationship for any kind of problem, the methodology adopted in this paper may be replicated for similar study.
- Based on the comparison of obtained R^2 values for linear and non-linear relationship, it can be concluded that mix-proportions and compressive strength have linear relationship.
- The results conclude that MOCS have a wider spread compared to NSGA-II and thereby state that MOCS is a good alternative Multi-objective optimization algorithm for solving such similar problems

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